Session 4 – Linear classifiers, data-dependent priors and synthesis

Goal. By the end of the session you should be able to implement PBGD, certify a Gaussian posterior exactly and by Monte Carlo, learn a prior on part of the data, and put all the results of the four sessions together.
To reread before the session. Sections 4.2, 4.3, 6.4 and 7.4 of the notes; notebook 5.
Timing (3 hours). Part 4.1 75 min
Part 4.2 50 min
Part 4.3 30 min
Part 4.4 25 min
To hand in at the end of the session. Your notebook, with the tables of Parts 4.1 and 4.2, the decision table of Part 4.3, and the outline of the final report.

Part 4.1 – (⋆) PAC-Bayesian learning of linear classifiers and Monte Carlo certificates (Sections 4.3 and 7.4)

Consider linear classifiers $h_{\mathbf{v}}(x) = \operatorname{sign}\langle \mathbf{v}, \tilde{x}\rangle$ on the standardized features augmented with a constant 1 ($\tilde{x} = (x, 1)$), the prior $P = \mathcal{N}(\mathbf{0}, I)$ and the posteriors $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I)$.

1. Implement the exact empirical Gibbs risk $\widehat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m}\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle / \|\tilde{x}_i\|)$ and the PBGD objective $C\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle / \|\tilde{x}_i\|) + \frac{1}{2}\|\mathbf{w}\|^2$ with its gradient; minimize it with L-BFGS, starting from a linear SVM.

2. On `australian`, `pima`, `wdbc`, `spambase`, compute Seeger's certificate on the Gibbs risk for 13 values of C in $[10^{-2}, 10^2]$, select the best one, and compare the test risk of the vote $\operatorname{sign}\langle \mathbf{w}, \tilde{x}\rangle$ with a linear SVM ($C = 1$). Plot the certificate as a function of C.

3. Pretend that the Gibbs risk has no closed form: estimate it with $N = 2000$ classifiers drawn from $Q_{\mathbf{w}}$ and compute the Monte Carlo certificate (Section 4.3), with $\delta/2$ for each step. How much is lost?

4. Why can C be selected by minimizing the certificate without any union bound?

Part 4.2 – (⋆) Data-dependent priors for linear classifiers (Sections 4.2 and 6.4)

1. Split the training set into S_1 (a fraction f) and S_2. Learn a prior centre $\mathbf{w}_0$ on S_1 (PBGD with $C = 1$), then run PBGD on S_2 with the prior $\mathcal{N}(\mathbf{w}_0, I)$, i.e. with the regularizer $\frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$, and compute Seeger's certificate on S_2 only ($\mathrm{KL} = \frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$), selecting C on the grid of Part 4.1.

2. Compare $f \in \{0, 0.2, 0.5\}$ ($f = 0$ is Part 4.1) on the four datasets: size of S_2, KL, certificate, test risk.

3. Which examples may the posterior use? Why is the certificate valid although $\mathbf{w}_0$ depends on the data?

Part 4.3 – Synthesis: which certificate for which situation?

Using your results of the four sessions (and not new computations), fill in the following decision table. For each situation, say which guarantee you would report (test-set bound, first-order, tandem or C-bound certificate, Seeger's bound on a Gibbs classifier, with or without data-dependent prior), and justify your choice in one sentence with a number from your experiments.

situation	guarantee to report, and why
a forest has been learned, and 30% of the data can be set aside	
a small dataset (a few hundred examples)	
the weights of the vote are learned from the data	
several models are compared and one is selected	
an imbalanced problem	
a boosted vote of stumps	
a stochastic (Gibbs) linear classifier	

Part 4.4 – Report (to finish after the session)

Summarize your results in a short report (3–4 pages) containing the tables of Parts 1.3 and 2.1, two figures of your choice and a discussion. As in the first practical session, compare methods that are comparable and give standard deviations; remember that a certificate is not an estimate of the risk, and comment on its tightness, not only on its value; finally, report every choice made using the data (grids, splits, selected bounds), since each of them may require a union bound.